Chapter 9c: Applications of Fixed Point Theorems

Integral and Integrodifferential Equations

Pathak's Nonlinear Analysis

July 14, 2026

Chapter 9: Applications of Fixed Point Theorems

- **9.1**: Existence of solutions to differential equations

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- **9.2**: Existence of solutions to differential inclusions

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- **9.4**: Applications to periodic problems
- **9.5**: Applications to nonlinear integral equations

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- **9.6**: Applications to abstract Volterra integrodifferential equations

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- **9.6**: Applications to abstract Volterra integrodifferential equations

Focus of this chapter:

- Existence and uniqueness theorems for integral equations
- Fixed point methods and contraction mappings
- Cone methods for positivity preservation
- Integrodifferential equations with operator semigroups

Key Concepts and Tools

Integral Equations:

- Volterra equations
- Fredholm equations
- Urysohn operators
- Generalizations

Fixed Point Methods:

- Contraction mappings
- Darbo's theorem
- Measure of noncompactness

Functional Spaces:

- $C[a, b]$ (continuous functions)
- L_p spaces
- Cone structures

Operators:

- Integral operators
- C_0-semigroups
- Infinitesimal generators

Volterra Integral Equations

Definition

A **Volterra integral equation** of the second kind is:

$$x(t) = \lambda \int_a^t f(t, s, x(s))ds + y(t), \quad t \in [a, b]$$

where λ is a parameter, f is a given function, and y is the inhomogeneous term.

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Theorem (Classical Contraction Solution)

If $|f(t, s, \xi) - f(t, s, \eta)| \leq F(t, s)|\xi - \eta|$ *with*

$$k = \sup_{t \in [a,b]} \left\{ \int_a^t F(t, s)ds \right\} < 1$$

then the Volterra equation has a unique solution in $C[a, b]$.

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Key observation: The contraction factor improves with the integral, making convergence fast.

Urysohn Operator

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$$[Ux](t) = \int_{\Omega} f(t, s, x(s))ds$$

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Theorem (Positivity in Cones)

Let $X = C[a, b]$, and K be the cone of nonnegative functions. If:

1. *U maps K into itself*
2. *U is continuous and maps bounded sets into precompact sets*
3. *Suitable boundary conditions hold*

then the equation $Ux = x$ has a positive solution.

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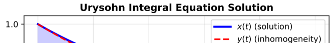

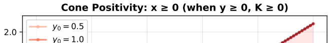

Theorem 9.26: Himmelberg's Lemma Application

Theorem

Let X be a separable reflexive Banach space and $C \subset X$ a closed convex subset. Suppose $\varphi_x : X \to Y$ is a continuous map for each $x \in C$ such that:

1. $(x_n \to x \text{ in } C, u_n \to u \text{ in } X) \Rightarrow \varphi_{x_n}(u_n) \to \varphi_x(u)$
2. *The set* $C_x = \{u \in X : \varphi_x(u) = 0\}$ *is nonempty, closed, and convex*
3. $C_x \cap B_r \neq \emptyset$ *for each* $x \in C$

then $\varphi_x(x) = 0$ *has a solution in* C.

Application to Integral Equations: Setting $\varphi_x(u) = u - Ux$ where U is an integral operator, we obtain solutions to nonlinear integral equations.

Theorem 9.27: Nonlinear Integral Equations

Theorem

Consider the integral equation:

$$x(s) = y(s) + \int_0^1 K(s,t)x(t)dt, \quad s \in [0,1]$$

Suppose:

1. $K(s,t) \geq 0$ *a.e. on* $[0,1] \times [0,1]$
2. $\operatorname{ess\,sup}_{t\in[0,1]} \int_0^1 K^2(s,t)dt < \infty$
3. $y(s) \geq 0$ *a.e. on* $[0,1]$

Then the equation has a solution $x \in L_2[0,1]$ *with* $0 \leq x(s) \leq y(s)$ *a.e.*

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Key idea: Using cone positivity and Himmelberg's lemma, we guarantee positive solutions under mild conditions on kernels.

Theorem 9.30: Periodic Solutions

Theorem

Consider the nonlinear integral equation:

$$x(t) = q(t) + \int_0^{\beta} f(t, s, x(\phi(s)))ds + \int_0^{\sigma(t)} g(t, s, x(\phi(s)))ds$$

Suppose certain conditions hold on f, g, and the domain $[0, \beta]$. Then the equation has a positive p-periodic solution defined on $\mathbb{R}$.

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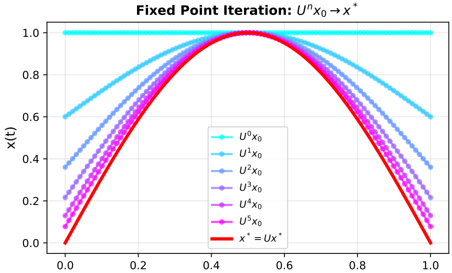

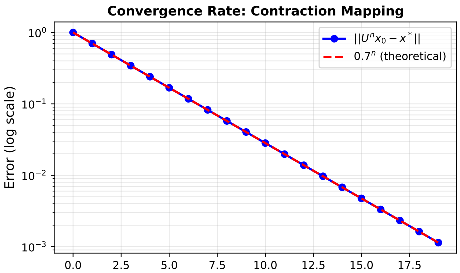

Theorem 9.32: Darbo Condition for Integral Operators

Theorem

Assume $\Omega \subset C[a,b]$ is nonempty, bounded, convex, and closed. Let P and T map Ω continuously into itself with $P(\Omega)$ and $T(\Omega)$ bounded. If P and T satisfy the Darbo condition with constants k_1, k_2:

$$\|P(\Omega)\|_{k2} + \|T(\Omega)\|_{k_1} < 1$$

then $S = P \circ T$ is a contraction with respect to the measure of noncompactness ω with constant $\|P(\Omega)\|_{k_2} + \|T(\Omega)\|_{k_1}$, and has at least one fixed point in Ω.

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Advantage over Banach: Darbo's theorem applies to non-contractive operators via the Hausdorff measure of noncompactness.

Example: Generalized Nonlinear Functional-Integral Equations

Consider the generalized nonlinear functional-integral equation (GNFIE):

$$x(t) = f\left(t, x(\theta(t)), \int_0^{\beta(t)} f(t,s,x(\phi(s)))ds, \int_0^{\sigma(t)} g(t,s,x(\phi(s)))ds\right)$$

Special cases:

When	**Reduces to**
$\sigma(t) = \infty$, $g = 0$	Nonlinear Fredholm IE (9.124)
$\sigma(t) = \infty$, $F(t,x) = f$	Integral equation (9.125)
$F(t,x) = f(t, x_2, x_4)$	Nonlinear IE (9.126)
$F = p(t, x_1) + x_2 x_4$	Quadratic integral IE (9.127)
$F = x_1 + p(x_3, x_4)$	Functional-integral IE (9.131)

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Each case can be studied separately with appropriate fixed point theorems!

Special Cases of Integral Equations

1. **Volterra IE (9.144):** $x(t) = a(t) + \int_0^t p(t, s, x(s))ds$
2. **Urysohn IE (9.145):** $x(t) = b(t) + \int_0^a q(t, s, x(s))ds$
3. **Functional IE (9.146):** $x(t) = a(t) \int_0^a q(t, s, x(s))ds + \left(\int_0^t p(t, s, x(s))ds\right) \left(\int_0^a q(t, s, x(s))ds\right)$
4. **Chandrasekhar IE (9.147):** $x(t) = 1 + x(t) \int_0^a \frac{t}{t+s}\varphi(s)x(s)ds$

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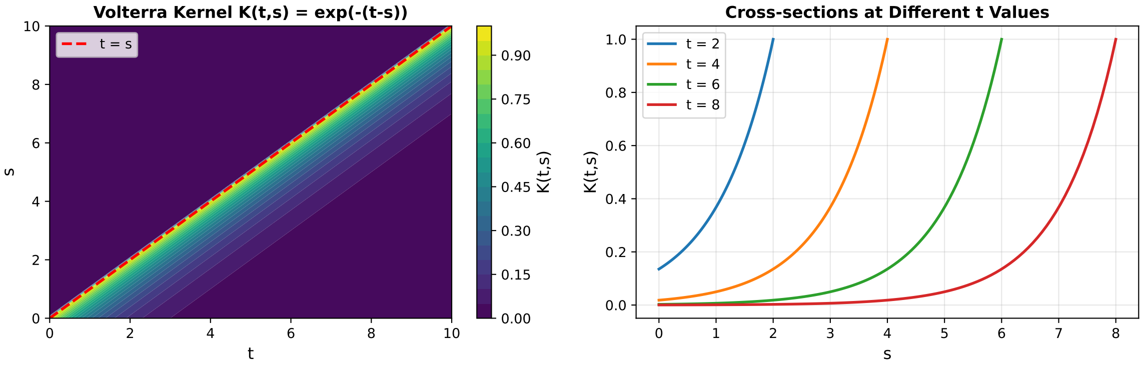
Common theme: All can be analyzed using fixed point theory via contraction properties or cone methods!

Numerical Example: Volterra Integral Equation

```python
# Solve: x(t) = sin(t) + 0.3*integral_0^t (t-s)*x(s) ds

import numpy as np
from scipy.integrate import odeint

def volterra_kernel(t, s):
    """Volterra kernel K(t,s) = t - s"""
    return max(t - s, 0)

def solve_volterra(kernel, y, t, iterations=5):
    """Fixed point iteration for Volterra equation"""
    x = y.copy()
    for k in range(iterations):
        x_new = np.zeros_like(x)
        for i, t_i in enumerate(t):
            # Numerical integration
            integral = 0
            for j in range(i):
                integral += 0.3 * kernel(t_i, t[j]) * x[j]
            x_new[i] = y[i] + integral
        x = x_new
    return x

# Parameters
t = np.linspace(0, 1, 100)
y = np.sin(t)
x_solution = solve_volterra(volterra_kernel, y, t)
```

Visualization of Volterra Kernel



The left plot shows the Volterra kernel $K(t,s) = e^{-(t-s)}$ as a contourf, while the right shows cross-sections at different t values. The causal structure $K(t,s) = 0$ for $t < s$ is essential for existence of solutions.

Abstract Volterra Integrodifferential Equations

Definition

An abstract Volterra integrodifferential equation of the form:

$$u'(t) + Au(t) = f(t, u(t)) + \int_{t_0}^{t} g\left(t, s, u(s), \int_{t_0}^{s} K_1(s, \tau, u(\tau))d\tau\right) ds$$

where:

- $-A$ is the infinitesimal generator of a C_0-semigroup $\{T(t)\}_{t\geq 0}$
- f, g are given continuous functions
- K_1, K_2 are integral kernels

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Mild solution: A continuous $u(t)$ is a mild solution if:

$$u(t) = T(t - t_0)u_0 + \int_{t_0}^{t} T(t - s)f(s, u(s))ds + \int_{t_0}^{t} T(t - s)\mathcal{K}(s)ds$$

where $\mathcal{K}(s)$ denotes the integral term.

Key Assumptions for IDEs

Hypothesis (H_1)**:** Functions u, f, g are continuous and bounded on relevant domains.

Hypothesis (H_2)**:** Lipschitz conditions:

$$|u(t, x_1) - u(t, x_2)| \leq a_1(t)|x_1 - x_2| \tag{1}$$
$$|f(t, y_1, x_1) - f(t, y_2, x_2)| \leq a_2(t)|y_1 - y_2| + a_3(t)|x_1 - x_2| \tag{2}$$
$$|g(t, y_1, x_1) - g(t, y_2, x_2)| \leq a_4(t)|y_1 - y_2| + a_5(t)|x_1 - x_2| \tag{3}$$

Hypothesis (H_3)**:** Continuous functions form the set $\mathcal{C} = [0, a] \times [0, a]$ mapping into $[0, a] \times \mathbb{R}$.

Hypothesis (H_4)**:** $\max\{a_1, a_2, a_3, a_4, a_5\}(t) \leq k$ (boundedness).

Hypothesis (H_6)**:** $4\gamma\eta < 1$ where $\gamma = k + ka\beta$, $\eta = kaa\ell + m$ (contraction condition).

Theorem 9.34: Common Mild Solution

Theorem

Consider the equations:

$$u'(t) + Au(t) = f(t, u(t)) + \int_{t_0}^{t} g\left(t, s, u(s), \int_{t_0}^{s} K_1(s, \tau, u(\tau))d\tau\right) ds \quad (4)$$

$$u'(t) + Au(t) = \bar{f}(t, u(t)) + \int_{t_0}^{t} \bar{g}\left(t, s, u(s), \int_{t_0}^{s} \bar{K}_1(s, \tau, u(\tau))d\tau\right) ds \quad (5)$$

Under hypotheses (H_1)-(H_6), these equations have a common mild solution in B_r.

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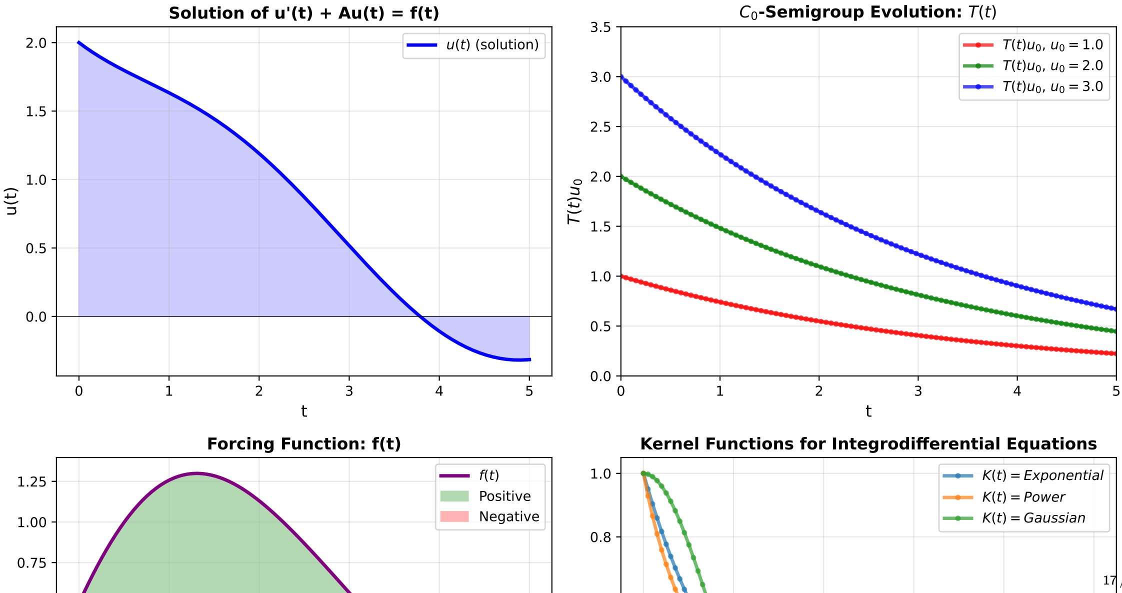
Proof technique: Define operators T_1, T_2 for each equation and use fixed point iteration on their composition.

Integrodifferential Equation Solution



Convergence Results

Theorem (Rates of Convergence)

For the contraction operator U with contraction constant $k < 1$, the iterates $U^n x_0$ converge to the fixed point x^* with:

$$\|U^n x_0 - x^*\| \leq \frac{k^n}{1-k}\|Ux_0 - x_0\|$$

The convergence is:

1. **Linear** *if k is constant*
2. **Superlinear** *if k depends on iteration and $k_n \to 0$*
3. **Quadratic** *if $k_n = O(\|U^n x_0 - x^*\|)$*

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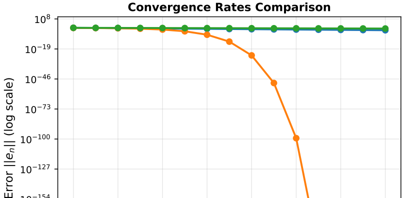

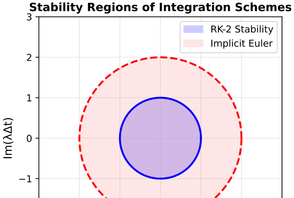

Stability Analysis

Theorem (Stability of Solutions)

If U is a contraction mapping on a Banach space X, then:

1. *The unique fixed point x^* is* **globally asymptotically stable**
2. *Small perturbations in initial data lead to small perturbations in solutions*
3. *The solution depends continuously on parameters*

For numerical schemes: The stability depends on the step size Δt:

- **Explicit Euler**: Stable if $\lambda \Delta t$ lies in unit disk
- **Implicit Euler**: Stable for all $\Delta t > 0$ (unconditionally stable)
- **RK-4**: Stable for $|\lambda \Delta t| \leq 2.78$ (larger region)

Measure of Noncompactness

Definition

The **Hausdorff measure of noncompactness** $\omega(A)$ of a bounded set A is:

$$\omega(A) = \inf\{d > 0 : A \text{ can be covered by finitely many sets of diameter} \leq d\}$$

Properties:

1. $\omega(A) = 0$ iff A is compact (or precompact)
2. $\omega(A) = \omega(\overline{A}) = \omega(\text{conv}(A))$
3. $\omega(A \cup B) = \max\{\omega(A), \omega(B)\}$
4. $\omega(A + B) \leq \omega(A) + \omega(B)$

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Key insight: Measures of noncompactness generalize compactness to infinite-dimensional spaces.

Darbo's Fixed Point Theorem

Theorem (Darbo, 1955)

Let K be a nonempty, bounded, convex, and closed subset of a Banach space X, and let $T : K \to K$ be a continuous operator. If there exists a constant $k \in [0, 1)$ such that:

$$\omega(T(A)) \leq k \cdot \omega(A)$$

for all bounded subsets $A \subseteq K$, then T has at least one fixed point in K.

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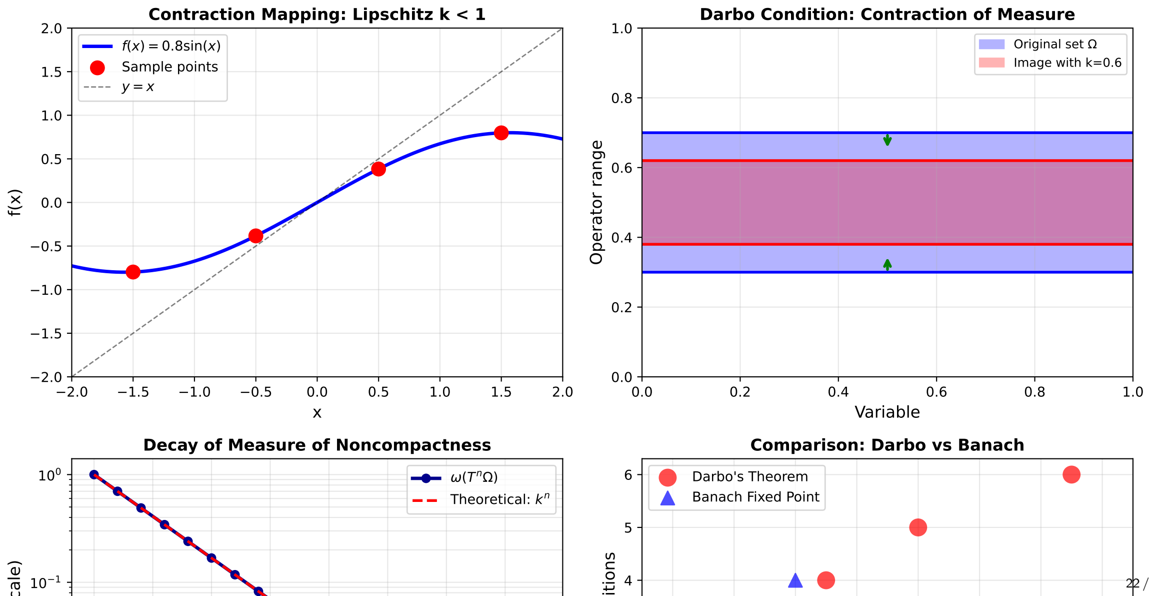
Advantages:

- Applies to operators that are **not contractive** in classical sense
- Works in spaces without metrics (only seminorms)
- Extends Banach fixed point theorem substantially

Darbo Condition Visualization



Main Results Summary

1. **Section 9.5:** Integral equations via fixed points
   - Volterra equations with contraction constant $k < 1$
   - Cone methods for positivity preservation
   - Generalized functional-integral equations
   - Multiple special cases (Fredholm, Urysohn, quadratic, etc.)

2. **Section 9.6:** Integrodifferential equations
   - Abstract equations with infinitesimal generators
   - C_0-semigroups and mild solutions
   - Common solutions to pairs of equations
   - Continuous dependence on data

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Unifying theme: Fixed point theory provides a powerful framework for proving existence and uniqueness of solutions to complex integral and integrodifferential equations.

Key Techniques

Functional Analysis:

- Banach spaces and norms
- Compactness and precompactness
- Measure of noncompactness
- Semigroup theory

Fixed Point Methods:

- Contraction mappings
- Darbo's theorem
- Himmelberg's lemma
- Krasnoselskii's theorem

Application Strategy:

1. Formulate the problem as $Tx = x$ for an appropriate operator T
2. Verify that T maps a suitable set into itself
3. Show T is continuous and maps bounded sets appropriately
4. Apply the relevant fixed point theorem

Implementation Considerations

When solving integral equations numerically:

1. **Discretization:** Use quadrature rules (Simpson, trapezoidal)
   - Higher order rules for smoother kernels
   - Special care near singularities
2. **Iteration:** Fixed point iteration usually stable
   - Gauss-Seidel faster than Jacobi
   - Damping helps: $x_{n+1} = \alpha T x_n + (1 - \alpha) x_n$
3. **Stopping criterion:** Use error estimates
   - A posteriori: $\|x_{n+1} - x_n\| < \epsilon$
   - Or check: $\|T x_n - x_n\| < \epsilon(1 - k)$

For integrodifferential equations: Use ODE solvers with semigroup evaluations.

Open Problems and Extensions

- **Nonlinear growth:** $|f(t,s,\xi)| \leq a(t,s) + b(t,s)|\xi|^p$
- **Delay arguments:** $x(t) = \int_0^t f(t,s,x(\theta(s)))ds$
- **Fractional equations:** $D^\alpha x = f(t,x) + \int_0^t (t-s)^\beta x(s)ds$
- **Coupled systems:** Multiple equations with coupling terms
- **Singular kernels:** $K(t,s) = (t-s)^\alpha$ with $\alpha < 0$
- **Random equations:** Stochastic integral equations with noise

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All extensions use similar fixed point methodology!

References and Further Reading

- **Pathak, H. K.** (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*
- **Gripenberg, G., Londen, S. O., & Staffans, O.** (1990). *Volterra Integral and Functional Equations*
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Thank You

Chapter 9c: Applications of Fixed Point Theorems to Integral and Integrodifferential Equations

For more information, see: *An Introduction to Nonlinear Analysis and Fixed Point Theory* (Pathak, 2018)